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Example: 12.20

A manufacturer estimates that, when units of a commodity are produced each month the total costs will be TC(Q) • 128 • 60Q • $8Q^2$ Find the marginal cost, average cost, fixed cost, variable cost, average fixed cost and average variable cost.

Solution:

Given that TC(Q) • 128 • 60Q • $8Q^2$

We know TC • Fixed cost • variable cost

$$MC\ (Q) \bullet \frac{d(TC)}{dQ}$$

$$\bullet\ 0 \bullet 60(1)Q^{1\bullet 1} \bullet 8(2)Q^{2\bullet 1}$$

$$\bullet\ 0 \bullet 60Q^{0} \bullet 16Q^{1}(\text{Since, } Q^{0}\bullet 1)$$

$$\boxed{MC \bullet 60 \bullet 16Q}$$

$$Average\,Cost = \frac{TC}{Q}$$

$$\bullet\bullet\frac{128+60Q+8Q^2}{Q}$$

$$AC \bullet\bullet\frac{128}{Q}+60+8Q^2$$

Constant value is known as fixed cost

Fixed cost • 128

FC • 128

$$\text{Average Fixed cost} \bullet\bullet\frac{128}{Q}$$

$$AFC \bullet \frac{128}{Q}$$

Average Variable cost • 60 • 8Q (variable cost divided by Q)

$$\therefore AVC \bullet 60 \bullet 8Q$$

12.4.8 Elasticity of Demand

Elasticity of Demand is the ratio of the proportionate change in quantity demanded to the proportionate change in price. In mathematical terms,

$$e_d \bullet \left(\frac{P}{x}\right)\left(\frac{dx}{dp}\right)$$

In demand function Q = a–bP

So, e = (dQ/dP)(P/Q)

Example 12.21

If the demand function is $x \bullet \frac{100}{P}$, find e_d with respect to price at the point where P • 2

Note

By taking supply function, the elasticity of supply can be calculated

Solution:

Given

$$x = \frac{100}{P} = 100P^{-1}$$

$$\therefore \frac{dx}{dp} = 100(-1)P^{-1-1}$$

$$= 100(-1)P^{-2}$$

$$\bullet\ \bullet\ 100(P^{\bullet 2})$$

$$= \frac{-100}{P^2}$$

$$\frac{dx}{dp} = \frac{100}{4} = 25 \text{ and } x = \frac{100}{2} = 50$$

At P• 2,

Substituting the values in formula

$$e_d = \frac{Pdx}{xdp} = \left(\frac{2}{50}\right)\left(\frac{-100}{4}\right) = \frac{-200}{200} = -1$$

$$e_d \bullet\ \bullet 1$$